SHANKARAN R

B.E. ELECTRONICS AND COMMUNICATION ENGINEERING

+91 90807 47664 | shankaranramesh96@gmail.com | Kallakurichi, Tamil Nadu

PROFESSIONAL SUMMARY

Motivated Electronics and Communication Engineering student with a knowledge of Embedded C, microcontrollers, sensors, and hardware interfacing. In C and Java with practical experience in developing embedded and IoT-based projects. Seeking an entry-level Embedded Systems Engineer role to apply technical knowledge and contribute to innovative product development.

TECHNICAL SKILLS

Programming: C, Embedded C, Java
Embedded Systems: Microcontrollers, GPIO, UART, ADC, Sensors, Hardware Interfacing
Hardware: PCB Basics, Electronic Sensors

INTERNSHIP EXPERIENCE

CODEBIND TECHNOLOGIES | Embedded Systems Intern | June 2025

• Gained hands-on experience in Embedded C programming and microcontroller-based application development.
• Worked with GPIO, UART, ADC, sensors, hardware interfacing, debugging, firmware development, and hardware-software integration.

PROJECTS

Real-Time ECG Signal Monitoring and Analysis

• Developed an IoT-based real-time ECG monitoring system using ESP32 and AD8232 ECG sensor.
• Acquired and processed ECG signals to monitor heart rate and identify abnormal conditions.
• Integrated GPS and emergency alerts for real-time monitoring and location-based assistance.

Fire and Gas Detection & Automatic Safety System

• Developed an automated safety system for real-time detection of fire, smoke, and gas leakage using sensors and a microcontroller.
• Implemented buzzer and alert mechanisms for hazardous conditions and automatic safety monitoring.

Design of Compact Planar Terahertz Microstrip Antenna for High-Speed Wireless Applications

• Designed and simulated a compact planar microstrip antenna for terahertz-frequency communication.
• Analyzed return loss, bandwidth, gain, and radiation characteristics for high-speed wireless applications.

EDUCATION

Qualification	Institution	Year	Score
B.E. Electronics and Communication Engineering	Kongunadu College of Engineering and Technology	2023–2027	CGPA: 7.0
HSC	AKT Academy Matric Hr. Sec. School	2023	70%
SSLC	AKT Academy Matric Hr. Sec. School	2021	Completed

CERTIFICATIONS

• NPTEL – Introduction to Internet of Things • NPTEL – Industry 4.0 • Core Java Certification • Embedded with AI Certification

ACTIVITIES & ADDITIONAL INFORMATION

• Participated in technical paper presentation events and college symposiums.
• Presented technical topics related to Embedded Systems and IoT.
Languages: Tamil, English | Hobby: Playing and Watching Cricket